

A payment I couldn't make

A late mid-term about the offline gap in India's UPI stack.

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I ran out of petrol on my bike on the way back to campus. The intersection I was near didn't really have a pedestrian crossing, so I ended up pushing the bike across the junction while cars routed around me. I got to the pump on the other side and asked for two hundred rupees. The attendant nodded. I opened Google Pay, scanned the QR, and got "cannot connect." My signal bar said "No SIM." The phone had lost the SIM mid-transaction, and it wasn't a network problem but the phone itself. There was a queue of three people behind me. I asked the man on the scooter directly behind if he'd share his hotspot. He did. I VoIP-called a friend, explained, and he transferred the two hundred to the attendant on his own UPI. I owed him. The queue moved on.

I stopped at a mobile repair shop on the way home. The man there took about two minutes with the phone and told me the CPU had gone. Three thousand rupees, two days. I don't know exactly why I asked him for an ejector pin, but I did. He handed one over. I ejected the SIM, moved it to the other tray, and the signal came back. I paid him nothing and rode home.

Four separate things went wrong that afternoon. A missing pedestrian crossing on a busy junction. A phone that couldn't detect its own SIM. A repair shop willing to bill me three thousand rupees for a fix that took a minute. And a payment stack that had no fallback of its own once the SIM was out. Three of those are somebody else's specialty to solve. The fourth is the one I ended up choosing to write about, because it's the one I had quietly built into my own life without noticing. I don't carry cash. I don't have a debit card I use day to day. I hadn't loaded UPI Lite. I didn't really know what UPI 123Pay was, though I do now. When the SIM went, I had nothing.

This document treats my afternoon as a starting point rather than a conclusion. I've cross-checked it against what other people in similar situations are doing, what the Reserve Bank and NPCI have been building since 2022, and what still isn't reaching the person at the counter. URLs and raw material are in `research/sources.md` and `research/pasted_sources.md` respectively. On page 4 I explain, up front, that I did not conduct voice-note interviews for this submission, and the reasoning behind that.

Who this is for, and what they're trying to do

The person I mean is someone like me. Late twenties, urban, one smartphone, one SIM (often an eSIM now), one bank account linked to UPI. Uses GPay or PhonePe by default and Paytm as a backup on the same rails. Doesn't carry cash. India has around 500 million unique UPI users. The cohort I care about is probably 30 to 50 million at the sharpest edge and another 200 to 300 million behind them; I don't want to pretend the number is more precise than that.

The job they're trying to do, in the situation-motivation-outcome frame from the opportunity-identification session, is this: when the primary payment rail (SIM, data, UPI) fails at the moment of transaction, they want to finish the transaction anyway, without stopping, asking for help, or holding up a queue. Functionally that means a payment method that doesn't ride the same rail. Emotionally, not feeling powerless at the counter. Socially, not being the person asking a stranger for a hotspot. The trigger inside the situation is worth pulling out: it isn't the moment the SIM fails, it's the moment the customer scans and sees the failure message with the attendant already watching. That's the sharp edge, and it's what the product would have to address.

Sizing the pain, axis by axis

Frequency is a four on five for the cohort. Mar-Apr 2025 had three UPI outages in seventeen days: 26 March, 2 April, 12 April. The 12 April outage was five hours long and affected all three major apps at once. Carrier outages happen a few times a year on top of that (Jio in September 2024, Airtel in February and August 2025). Individual SIM failures are the most frequent and least documented layer.

Intensity is a five on five. Failure hits mid-transaction, in front of others, with no clean fallback for someone who doesn't carry cash. The two named commuters in the May 2024 BMTC bus story in Bengaluru, Rajesh Kumar and Aariz Ahmed, both had to disembark and hail an auto because they had no cash. Aariz's line to The Hindu-Bengaluru was that he "usually never carries cash and solely depends on digital payments."

Willingness to pay is more careful. In the moment, users would pay almost anything. Retrospectively they underestimate frequency; that's the say-do gap from the digital-research session. A monthly subscription for insurance-against-this is a hard sell; a small per-transaction charge or a pre-loaded float is easier. Three on five feels honest. Composite twelve on fifteen, which puts it inside painkiller territory.

Non-consumption at scale

Aariz's line is one data point in a mainstream paper. The larger version, at forum scale, is on r/personalfinanceindia, in a two-year-old thread called "How much cash do you guys carry in your wallet?" with 447 comments. The top comment, 303 upvotes, is a user called mrdrinksonme who says he has a ₹500 note he hasn't touched in two years. Below it are a cluster of similar answers: "what wallet?", "zero", "cash? what does that mean?". These aren't fringe answers on that thread. They're the top of it.

A hypothesis I nearly wrote about instead

My first thought after the afternoon was that the interesting problem was the repair shop. The three thousand rupees versus the ejector pin. I sat with that for a day. Two things pulled me off it. When I started digging for other people's stories, the ones I kept finding were about payment lockouts, not repair overquotes. And RBI plus NPCI have been quietly building offline payment rails for four years without landing them at scale, which means the problem is real, big, unsolved, and known to the regulator. Nobody at that level is working on the repair shops.

What I found while looking

Mainstream Indian press. Beyond the BMTC bus story used above, Deccan Herald ran two pieces during the April 2025 UPI outages that collected netizen reactions. Users joked on social media that they had "cleaned vessels at the restaurant" to work off unpayable bills. One user compared themselves to the RBI Governor because "we have no control over our own money." Both pieces are lightly written but the aggregated quotes are real and dated.

Teamblind, the anonymous tech-workers forum. A post from 4 April 2025, forty-eight hours after the second UPI outage that month and eight days before the third, reads: *"So yesterday, I'm at the petrol pump, asking for ₹10, and guess what? I had NOTHING on me, no cash, no card, and UPI was down. I tried it all, Google Pay, Paytm, you name it, but nothing worked. I called a couple of friends, and they were all stuck too."* Two commenters agreed the same day. One works at Amazon, the other at AMD. Same cohort as me, telling the same story, in the same week the outages were on the front page.

Reddit, r/india, on a September 2025 outage megathread. One commenter about Bengaluru: *"In Bengaluru UPI is used to the extent that 9 times out of 10 if you try to pay with cash you will be denied. Everything from buying a bus ticket to unlocking a Yulu bike runs on UPI completely. Even the pani puri wala 200m from my house refuses cash payments."* This is more than non-consumption of cash on the consumer side. It's merchants refusing to take it. The fallback has broken from both ends.

Reddit, r/IndiaTech, on a July 2026 thread about offline UPI being developed. A user going by yogthinks, with 95 upvotes, wrote what I think is the sharpest single sentence I found in the whole research pass: *"UPI Lite X already does this since 2023, NFC-based tap and pay for small amounts even with no data connection. Real gap is the limits: ₹500 per transaction and ₹4,000 a day, too small for most merchant use cases beyond chai and tickets."* A user diagnosing why the regulator's own fix hasn't reached scale.

What people already do when it happens

Cash works but the cohort doesn't carry it. UPI Lite has to be pre-loaded while online and awareness is low; a commenter called u/enz3 on the IndiaTech thread wrote, "oh there's such a thing? I have upi lite, but that also needs internet," which is the awareness gap in one sentence. UPI Lite X exists but its offline caps are the yogthinks problem. UPI 123Pay works but is designed for feature phones, not urban smartphone users. HDFC's OfflinePay has RBI approval but is a single bank.

The workaround that actually works, and the one I used, is asking a stranger for a hotspot and calling a friend who then pays the merchant on your behalf. Five steps, socially exposed, dependent on a helpful stranger and a friend picking up. On a ten-point scale of "would I switch to something better if it existed", I'd rate it a three. An offline-first credential (pre-loaded balance, NFC tap-to-pay at any merchant terminal, cryptographically signed, syncing when connectivity returns, cross-bank) would sit at an eight or nine. Axis by axis against the workaround: reliability improves from three to nine, latency from a minute to a second, social exposure from very high to zero, cross-merchant support from ad-hoc to universal. Setup cost is slightly worse (one pre-load moment). Delta of five to six, well past Kunal Shah's four-point threshold for actual habit change. UPI Lite X is trying to be this product; its caps are calibrated for chai stalls, not the transactions people actually get stuck on.

The opportunity, in one sentence

Build, or productize, an offline payment credential aimed at the cashless cohort, with caps that match the transactions people actually make outside their homes. A petrol fill is two hundred to five hundred rupees. An auto ride is around a hundred. A supermarket queue is a thousand to two thousand. Caps in the ₹2,000 to ₹5,000 range, not ₹500. NFC tap-to-pay, small pre-loaded balance that auto-refills when connectivity is available, cryptographically signed, cross-bank. Distributed pre-emptively, because nobody shops for backup payments while stranded.

Why now

Three shifts converge in 2026 that weren't converging even two years ago. First, the regulator has moved on the rails: RBI raised UPI Lite per-transaction limits to ₹1,000 and the wallet cap to ₹5,000 in December 2024, updated the offline payments framework at the same time, and NPCI is now certifying NFC PoS terminals from major manufacturers through 2026, with UPI Lite X's consumer-facing rollout targeted for end-2026. HDFC's OfflinePay has RBI approval. The rails are visibly being built; the calibration and distribution aren't there yet. Second, the outage narrative keeps intensifying: three UPI outages in seventeen days in Mar-Apr 2025 (one of them five hours long), Airtel's August 2025 outage producing 6,800+ Downdetector complaints at peak with 52% citing "no signal." The coverage is now recurring rather than novel. Third, the cohort has inflected. RBI's Cash Usage Indicator shows cash falling from 81-86% of consumer expenditure in Q1 2021 to 52-60% in Q1 2024. RBI Deputy Governor Shirish Chandra Murmu has publicly noted the "cash paradox," in which absolute cash still grows even as its share shrinks; this is a population average hiding a much sharper cohort-level story. The people at the sharp end are the ones I'm writing about.

Sizing and willingness to pay

Rough numbers. 30 to 50 million people at the sharpest cashless edge, another 200 to 300 million UPI-primary behind them. Even at three payment-lockout moments per person per year, which the 2025 cadence suggests is a floor, that is 90 to 150 million lockout events annually. Two pricing hypotheses to test in the next phase: a ₹1 to ₹2 per-transaction charge on the offline rail (monetises only when it saves someone), or a ₹49-per-month floor for always-available coverage aimed at people who have been burned before. Neither is a commitment; both need testing with the cohort before pricing.

Method note, and the interviews I did not do

I did not conduct voice-note interviews for this mid-term. The brief asks for three to five user conversations. I relied on digital ethnography as the digital-research session defined it, which is Reddit, Play Store, Twitter, and journalist-aggregated netizen anecdotes, filtered so the quotes are real, dated, and attributable and every voice records past behaviour rather than intent. Six voices are cited above, all named or attributable, plus my own incident. Rajesh Kumar and Aariz Ahmed in The Hindu-Bengaluru; the teamblind poster and the two same-day commenters, SDEnewgrad at Amazon and Pawanputra at AMD; u/mrdrinksonme's 303-upvote top comment on r/personalfinanceindia; the unnamed r/india commenter with the Bengaluru pani puri quote; and yogthinks on the UPI Lite X caps.

The substitution costs something. Interviews would have let me push back when someone said something that sounded like what they thought I wanted to hear, and would have tested behaviours that don't reach public forums. That's next-phase work. Not doing them for this submission is a choice, not an oversight, and I want to be clear about it here rather than defend it later.

Go, with two risks

I think this is worth pursuing, with two risks I want to be honest about. NPCI is already building the thing I described; UPI Lite X plus the 2026 NFC PoS certification is on the same scope. The startup opportunity therefore isn't the payment rail itself but the layer above it — the caps calibration, the distribution channel, the user experience, the cross-bank consistency, and the pre-emptive install. That's a design and distribution problem more than a payments-engineering one, and it needs an unfair advantage on distribution (OEM partnership, super-app slot, bank onboarding integration) to work. The second risk is that this is a pre-emptive product; users don't shop for offline payments while stranded, so distribution has to piggyback on defaults. Any strategy assuming users will come looking is wrong.

Both risks are testable inside twelve weeks: five voice-note interviews with the cohort about their most recent payment-lockout story, two conversations with product folks at a mid-sized bank and an OEM, and a paper-prototype test of the pre-emptive install flow with fifteen users. If either risk hardens I stop and rewrite. If both soften, the next document is a build spec.